STEP-BY-STEP PRACTICAL BLUEPRINT

Power BI Visuals Creation & Architecture Manual

Exact click actions, field bucket mappings, and annotated visual anatomy using your generated workbook dataset.

Prepared for & Authored by: Rupankar Ghosh

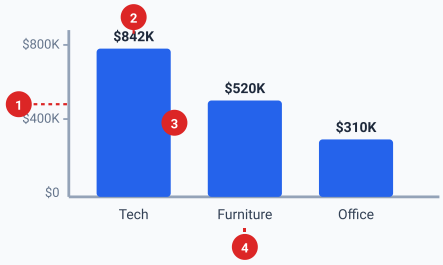
1. Categorical & Volume Foundations (Tier 1)

1.1 Clustered Column Chart

Field Bucket	Exact Field to Drag	Action / Aggregation
X-Axis	`Fact_Sales[Category]`	Categorical grouping (Labels)
Y-Axis	`Fact_Sales[Revenue]`	Numeric measure (Sum)
Legend (Optional)	`Fact_Sales[Region]`	Creates side-by-side grouped bars
Tooltips	`Fact_Sales[Profit Margin %]`	Average aggregation

1. Click **Clustered Column Chart** icon from the *Visualizations Pane*.
2. Drag `Category` into **X-axis**; drag `Revenue` into **Y-axis**.
3. Go to **Format Visual > Data labels > Toggle ON** to show dollar values above bars.
4. Under **Y-axis** formatting, set *Display Units* to "Thousands ($K)" for clean scale.

ANNOTATED VISUAL DIAGRAM

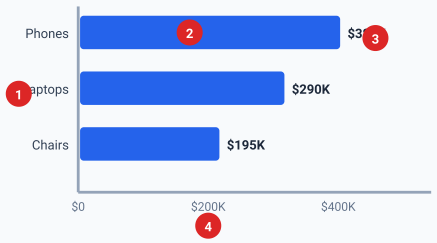


1 Y-Axis Metric (Sum of Revenue)
2 Data Labels (Turned On) 3 Column Bar Series
4 X-Axis Categories

1.2 Clustered Bar Chart (Horizontal)

Field Bucket	Exact Field to Drag	Action / Aggregation
Y-Axis	`Fact_Sales[Subcategory]`	Categorical items (Long text labels)
X-Axis	`Fact_Sales[Revenue]`	Quantitative bar length (Sum)
Legend	`Fact_Sales[Segment]`	Segment color breakdown

ANNOTATED VISUAL DIAGRAM



1 Y-Axis Category Labels 2 Horizontal Bar Length
3 Outside Data Label 4 X-Axis Numeric Measure Scale

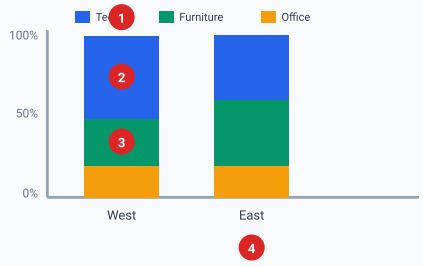
1. Select **Clustered Bar Chart** (horizontal icon) for readability when labels are long.
2. Drag `Subcategory` to **Y-axis** and `Revenue` to **X-axis**.
3. In visual header menu, click **Sort axis > Sort descending by Revenue**.
4. Enable **Data labels > Position: Outside End** to keep bars uncluttered.

1.3 Stacked & 100% Stacked Column Chart

Field Bucket	Exact Field to Drag	Action / Aggregation
X-Axis	`Fact_Sales[Region]`	Primary column grouping
Y-Axis	`Fact_Sales[Revenue]`	Cumulative height (100% normalized in 100% stack)
Legend	`Fact_Sales[Category]`	Segments within each column

1. Select **100% Stacked Column Chart** for percentage share comparisons.
2. Drag `Region` to **X-axis**, `Revenue` to **Y-axis**, and `Category` to **Legend**.
3. Go to **Format > Data labels > Display units > None**, toggle **Percent of total**.
4. Legend colors assign automatically to Technology, Furniture, and Office Supplies.

ANNOTATED VISUAL DIAGRAM



1 Category Legend Key 2 Segment Segment Share %
3 Middle Stack Partition 4 Categorical X-Axis Group

1.4 Line & Area Chart (Time-Series)

Field Bucket	Exact Field to Drag	Action / Aggregation
X-Axis	`Dim_Calendar[Date]` or `[Month_Name]`	Continuous chronology
Y-Axis	`Fact_Sales[Revenue]`	Numeric volume (Sum)
Secondary Y-Axis	`Fact_Sales[Profit Margin %]`	Dual-axis tracking (Average)
Legend	`Fact_Sales[Region]`	Generates multi-line series

ANNOTATED VISUAL DIAGRAM



1 Peak Data Point 2 Area Shading (Cumulative Volume)
3 Data Marker (Individual Points)
4 X-Axis Continuous Time Dimension

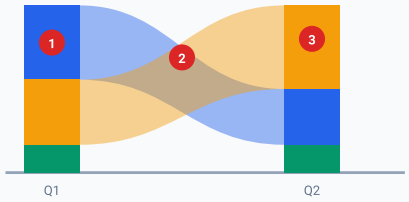
1. Select **Line Chart** or **Area Chart** from the palette.
2. Drag `Date` into **X-axis**; verify it is set to Date (not Hierarchy) for continuous line.
3. Under **Format Visual > Lines > Stroke width**, set to 3px and toggle **Markers ON**.
4. Go to **Analytics Pane** (magnifying glass) > Add **Trend Line** or **Forecast**.

1.5 Ribbon Chart (Rank Dynamics)

Field Bucket	Exact Field to Drag	Action / Aggregation
X-Axis	`Dim_Calendar[Quarter]`	Time periods (Q1, Q2, Q3, Q4)
Y-Axis	`Fact_Sales[Revenue]`	Ranking metric (Sum)
Legend	`Fact_Sales[Category]`	Ribbon ribbons competing for rank #1

1. Click **Ribbon Chart** icon.
2. Drag `Quarter` to **X-axis**, `Revenue` to **Y-axis**, and `Category` to **Legend**.
3. Observe how ribbons cross over one another as category ranks swap places.
4. Hover over ribbon connection bands to view *Rank Change* and *Growth Difference* in the tooltip.

ANNOTATED VISUAL DIAGRAM



1. Period 1 Rank #1 Leader
2. Ribbon Crossover (Rank Swap Bridge)
3. Period 2 New Rank #1 Leader

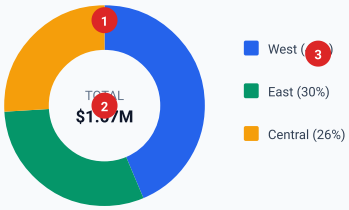
2. Part-to-Whole & Composition Visuals

2.1 Donut Chart

Field Bucket	Exact Field to Drag	Action / Aggregation
Legend	`Fact_Sales[Region]`	Segment slices (Max 3-5 categories)
Values	`Fact_Sales[Revenue]`	Arc circumference proportional weight
Tooltips	`Fact_Sales[Profit Margin %]`	Additional context on hover

1. Select **Donut Chart** from the gallery (preferred over Pie for readability).
2. Drag `Region` to **Legend** and `Revenue` to **Values**.
3. In **Format Visual > Slices > Inner radius**, tune the hollow core size (50-60%).
4. Under **Detail labels**, set Label Contents to "Category, percent of total".

ANNOTATED VISUAL DIAGRAM



1. Arc Slice (Proportional Revenue)
2. Inner Hollow Center Core
3. Category Legend & Percent Share

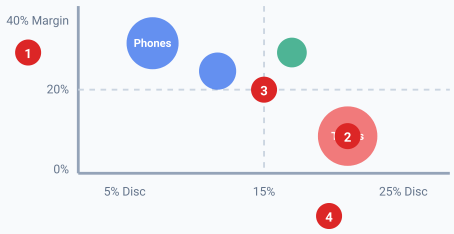
3. Correlation & Distribution Visuals

3.1 Scatter & Bubble Chart (Correlation)

Field Bucket	Exact Field to Drag	Action / Aggregation
Values	`Fact_Sales[Subcategory]`	Individual dot representation
X-Axis	`Fact_Sales[Discount_Rate]`	Independent variable (Average)
Y-Axis	`Fact_Sales[Profit Margin %]`	Dependent variable (Average)
Size (Bubble)	`Fact_Sales[Revenue]`	Bubble diameter scale (Sum)
Legend	`Fact_Sales[Category]`	Bubble color clustering

1. Click **Scatter Chart** icon.
2. Place `Subcategory` in **Values**, `Discount_Rate` on **X-axis**, and `Profit Margin %` on **Y-axis**.
3. Drag `Revenue` into **Size** to transform it into a 3-dimensional bubble chart.
4. In **Analytics Pane**, add **Trendline** and **Median X/Y Lines** for quadrant analysis.

ANNOTATED VISUAL DIAGRAM



1. Y-Axis Dependent Variable (Profit Margin %)
2. Bubble Size Encoding (Revenue Volume)
3. Median Quadrant Crosshair
4. X-Axis Independent Variable (Discount Rate)

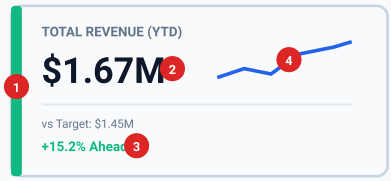
4. KPIs, Gauges & Target Tracking

4.1 Card (New) Visual & Reference Labels

Field Bucket	Exact Field to Drag	Action / Aggregation
Data	`Fact_Sales[Revenue]`	Primary callout big metric (Sum)
Reference Labels	`Fact_Targets[Target_Revenue]`	Baseline benchmark reference
Sparkline (New)	`Dim_Calendar[Date]`	Mini trend embedded inside card

1. Select **Card (New)** visual (contains lightning bolt icon).
2. Drag `Revenue` into **Data** field bucket.
3. Under **Format > Reference labels > Add Data**, drag `Target_Revenue`.
4. In **Accent bar**, toggle ON with green color (#10b981) for positive status.

ANNOTATED VISUAL DIAGRAM



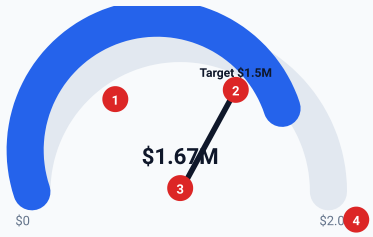
1. Status Accent Bar (Green / Red)
2. Big Callout Number (Primary Metric)
3. Reference Label (Target Comparison)
4. Embedded Context Sparkline

4.2 Radial Gauge Visual

Field Bucket	Exact Field to Drag	Action / Aggregation
Value	`Fact_Sales[Revenue]`	Actual current result (Fill arc)
Minimum value	Constant or `$0`	Base starting point of gauge
Maximum value	`Fact_Targets[Stretch_Goal]`	Top dial ceiling (e.g. $2.0M)
Target value	`Fact_Targets[Target_Revenue]`	Needle line indicator ($1.5M)

1. Select **Gauge** visual from the visualization pane.
2. Drag `Revenue` to **Value**, and `Target_Revenue` to **Target value**.
3. In **Format Visual > Gauge axis**, set Min: 0, Max: 2,000,000.
4. Under **Colors**, set Target needle color to dark navy (#0f172a).

ANNOTATED VISUAL DIAGRAM



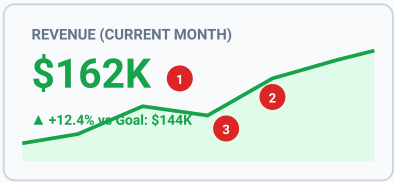
1. Filled Arc (Actual Revenue)
2. Target Needle Line Marker
3. Central Big Number Callout
4. Axis Maximum Ceiling Value

4.3 Power BI Standard KPI Visual

Field Bucket	Exact Field to Drag	Action / Aggregation
Value	`Fact_Sales[Revenue]`	Active measurement result (Sum)
Trend axis	`Dim_Calendar[Month_Name]`	Background silhouette area trend
Target	`Fact_Targets[Target_Revenue]`	Benchmark comparison target

1. Select **KPI** visual from the visualization tray.
2. Drag `Revenue` to **Value**, `Month_Name` to **Trend axis**, and `Target_Revenue` to **Target**.
3. In **Format Visual > Color coding**, set Direction = High is Good.
4. Status color automatically turns Green when Actual >= Target, Red when below.

ANNOTATED VISUAL DIAGRAM



1. Status Value (Green=Ahead, Red=Behind)
2. Background Trend Silhouette
3. Target Variance & % Delta

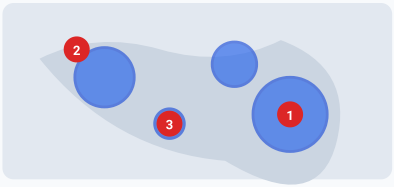
5. Geospatial & Map Visuals

5.1 Map (Bubble GPS) & Filled Map (Choropleth)

Field Bucket	Exact Field to Drag	Action / Aggregation
Location	`Fact_Sales[State]` or `[Country]`	Standard geographic boundaries
Latitude / Longitude	`Fact_Sales[Lat]`, `[Long]`	Exact precision pinpoint coordinates
Bubble size	`Fact_Sales[Revenue]`	Scale bubble diameter proportionally
Category / Legend	`Fact_Sales[Region]`	Pins colored by sales territory

1. Select **Map** (Bubble) or **Filled Map** (Choropleth polygon) visual.
2. Drag `Lat` and `Long` into their respective coordinate field wells.
3. Add `Revenue` into **Bubble size** to display transactional volume.
4. In **Format > Map settings > Style**, choose *Light* or *Grayscale* for executive reports.

ANNOTATED VISUAL DIAGRAM



1. High-Volume Store Bubble
2. Choropleth Geographic Territory
3. Low-Volume Regional Pin

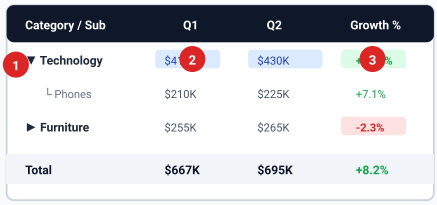
6. Tabular & Interactive Matrix Visuals

6.1 Matrix Visual with Data Bars & Heatmaps

Field Bucket	Exact Field to Drag	Action / Aggregation
Rows	`Fact_Sales[Category]`, `[Subcategory]`	Hierarchical row groupings with (+) drill
Columns	`Dim_Calendar[Quarter]`	Horizontal matrix columns (Q1-Q4)
Values	`Fact_Sales[Revenue]`, `[Profit Margin %]`	Aggregated summary cells

1. Select **Matrix** visual (not the plain Table).
2. Drag `Category` into **Rows** and `Quarter` into **Columns**.
3. Right click `Revenue` under Values > **Conditional formatting > Data bars > Enable**.
4. Right click `Profit Margin %` > **Background color > Color scale** (Red to Green).

ANNOTATED VISUAL DIAGRAM



1. Row Hierarchy Toggle (+/- Expand)
2. In-Cell Data Bar Conditional Format
3. Background Heatmap Color Scale

7. Advanced AI & Machine Learning Visuals

7.1 Key Influencers (ML Regression)

Field Bucket	Exact Field to Drag	Action / Aggregation
Analyze	`Fact_Customer_Churn[Churned]`	Target metric to explain (e.g. "Yes")
Explain by	`[Contract_Type]`, `[Monthly_Charges]`, `[Support_Tickets]`	Candidate driver attributes
Expand by (Opt)	`Fact_Customer_Churn[CustomerID]`	Row-level granularity

1. Click **Key Influencers** icon (sparkle bulb).
2. Set **Analyze** to `Churned` > select "What influences Churned to be Yes".
3. Add explanatory columns into **Explain by**.
4. Switch to the **Top Segments** tab to view clustered sub-populations with extreme churn rates.

ANNOTATED VISUAL DIAGRAM



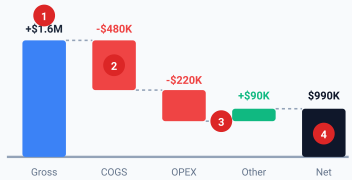
1. ML Factor Multiplier Effect
2. Comparative Probability Bar (Target vs Baseline)

7.2 Financial Waterfall Visual (Bridge)

Field Bucket	Exact Field to Drag	Action / Aggregation
Category	`Fact_Financial_Waterfall[Element]`	Stages (Gross, COGS, Opex)
Y-Axis	`Fact_Financial_Waterfall[Impact]`	Variance values (+/-)
Breakdown (Opt)	`Fact_Sales[Region]`	Splits each bar into regional drivers

1. Select **Waterfall Chart** visual from the gallery.
2. Drag `Financial_Element` to **Category**; drag `Impact_Amount` to **Y-axis**.
3. Go to **Format Visual > Columns > Colors**: verify Increase = Green, Decrease = Red, Total = Slate.
4. Check **Data labels** to verify sign polarity (+/- impact values).

ANNOTATED VISUAL DIAGRAM



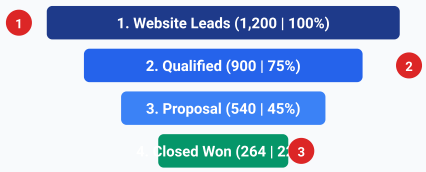
1. Starting Base Column
2. Negative Variance (Red Deduction)
3. Connector Bridge Line
4. Ending Net Profit Summary

7.3 Pipeline Funnel Visual

Field Bucket	Exact Field to Drag	Action / Aggregation
Category	`Fact_Sales_Pipeline[Stage]`	Ordered progression steps
Values	`Fact_Sales_Pipeline[Lead_Count]`	Count / Metric magnitude
Tooltips	`Fact_Sales_Pipeline[Conversion]`	Stage-to-stage retention %

1. Click **Funnel Chart** icon.
2. Drag `Pipeline_Stage` into **Category**; drag `Lead_Count` into **Values**.
3. In *Data View*, sort `Pipeline_Stage` by `Stage_Order` column to enforce logical sequence.
4. Under **Format > Data labels**, set label style to "Value and % of first".

ANNOTATED VISUAL DIAGRAM



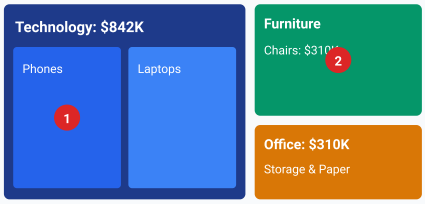
1. Ordered Stage Category
2. Funnel Width (Absolute Volume)
3. Final Retention % from Top

7.4 Hierarchical Treemap

Field Bucket	Exact Field to Drag	Action / Aggregation
Category	`Fact_Sales[Category]`	Top primary nested rectangles
Details	`Fact_Sales[Subcategory]`	Sub-tile partitions within group
Values	`Fact_Sales[Revenue]`	Sizes the rectangle proportionally

1. Insert **Treemap** from visual gallery.
2. Add `Category` into **Category**, `Subcategory` into **Details**.
3. Drag `Revenue` into **Values**.
4. Use the header drill-down arrows (drill down one level) to navigate hierarchical tiers.

ANNOTATED VISUAL DIAGRAM



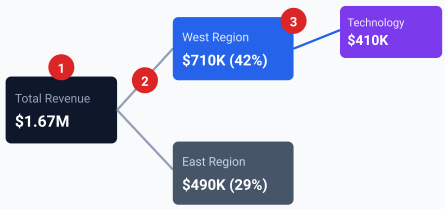
1. Subcategory Details Tile
2. Primary Parent Category Rectangle

7.5 AI Decomposition Tree

Field Bucket	Exact Field to Drag	Action / Aggregation
Analyze	`Fact_Sales[Revenue]`	Root metric to break down
Explain By	`Fact_Sales[Region]`, `[Category]`, `[Ship_Mode]`	Interactive candidate drill dimensions

1. Insert **Decomposition Tree** visual from the AI visual tier.
2. Drag `Revenue` to **Analyze**; drag dimensions to **Explain By**.
3. Click the **(+) icon** on the canvas and choose **High Value** to trigger AI pathing.
4. Notice the lightbulb icon signifying AI determination of maximal variance.

ANNOTATED VISUAL DIAGRAM



1. Root Analyze Metric
2. (+) Split Decision Selector
3. AI High-Value Drill Path